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COOLING SYSTEM

Abstract

A cooling system includes a first liquid cooling plate, a second liquid cooling plate, a first thermosiphon device, and a second thermosiphon device. The first liquid cooling plate has a first parallel flow channel formed by two flow channels connected in parallel. The second liquid cooling plate also has a second parallel flow channel formed by two flow channels connected in parallel. The first parallel flow channel and the second parallel flow channel are connected in series. The first thermosiphon device and the second thermosiphon device are both thermally coupled to the first liquid cooling plate and the second liquid cooling plate, and the first thermosiphon device and the second thermosiphon device are located between the first liquid cooling plate and the second liquid cooling plate.

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Background/Summary

BACKGROUND OF THE INVENTION

1. Field of the Invention

[0001] The present invention relates to a cooling system, and more particularly to a cooling system utilizing thermosiphon.

2. Description of the Prior Art

[0002] For a 1U water-cooled thermosiphon, the goals of low leakage hazard and high heat dissipation capability are achieved by combining the thermosiphon with a water cooling plate. In addition, the fins on the bottom of the water cooling plate can be used to cool the outlet airflow of the server, thereby reducing the load of the computer room air conditioner and improving cooling efficiency. However, the height difference between the evaporator and the condenser in the 1U space is limited. A smaller height difference will reduce the return capacity of the cooled water. In order to increase the height difference between the two as much as possible, the condenser can only be set at the highest position of the system. This configuration causes the condenser to be in contact with the water cooling plate only through its lower surface, causing uneven temperature of the condenser, reducing the capability of the thermosiphon to transfer heat to the water cooling plate, and limiting the cooling capacity of this cooling module.

SUMMARY OF THE INVENTION

[0003] An objective of the invention is to provide a cooling system, which uses two connected liquid cooling plates sandwiching two thermosiphon devices to improve the cooling capacity of the thermosiphon devices to heat sources.

[0004] A cooling system of an embodiment according to the invention includes a first liquid cooling plate, a second liquid cooling plate, a first thermosiphon device, and a second thermosiphon device. The first liquid cooling plate has a first left side flow channel and a first right side flow channel. The first left side flow channel and the first right side flow channel are connected in parallel to form a first parallel flow channel. The first liquid cooling plate and the second liquid cooling plate are spaced apart and opposite to each other. The second liquid cooling plate has a second left side flow channel and a second right side flow channel. The second left side flow channel and the second right side flow channel are connected in parallel to form a second parallel flow channel. The first parallel flow channel and the second parallel flow channel are connected in series. The first thermosiphon device is located between the first left side flow channel and the second left side flow channel and is thermally coupled to the first liquid cooling plate and the second liquid cooling plate. The second thermosiphon device is located between the first right side flow channel and the second right side flow channel and is thermally coupled to the first liquid cooling plate and the second liquid cooling plate. Thereby, compared with the configuration in the prior art that only one side of the condenser exchanges heat with the liquid cooling plate, both sides of the first and second thermosiphon devices (or the condensers thereof) can exchange heat with the first and second liquid cooling plates, so that the cooling capacity to the heat sources can be improved.

[0005] These and other objectives of the present invention will no doubt become obvious to those

of ordinary skill in the art after detailed description of the preferred reading the following embodiment that is illustrated in the various figures and drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- [0006] FIG. 1 is a schematic diagram illustrating a cooling system according to an embodiment.
[0007] FIG. 2 is a partially-exploded view of the cooling system in FIG. 1.
[0008] FIG. 3 is another partially-exploded view of the cooling system in FIG. 1.
[0009] FIG. 4 is a schematic diagram illustrating flow channels inside the first liquid cooling plate of the cooling system in FIG. 1.
[0010] FIG. 5 is a schematic diagram illustrating flow channels inside the second liquid cooling plate of the cooling system in FIG. 1.
[0011] FIG. 6 is a schematic diagram illustrating a plurality of columnar fins.
[0012] FIG. 7 is a schematic diagram illustrating a plurality of conical fins.
[0013] FIG. 8 is a schematic diagram illustrating a plurality of columnar fins with straight cuts.
[0014] FIG. 9 is a schematic diagram of a structure of fins formed by a plurality of sheets.

DETAILED DESCRIPTION

- [0015] Please refer to FIG. 1 to FIG. 3. A cooling system 1 according to an embodiment includes a first liquid cooling plate 12, a second liquid cooling plate 14, a first thermosiphon device 16, and a second thermosiphon device 18. The first liquid cooling plate 12 and the second liquid cooling plate 14 are spaced apart and opposite to each other in a first direction D1 (indicated by a double-headed arrow in the figures). Their working fluid flow channels communicate. A surface 12a of the first liquid cooling plate 12 is opposite to a surface 14a of the second liquid cooling plate 14. The first thermosiphon device 16 and the second thermosiphon device 18 are sandwiched by the first liquid cooling plate 12 (or the surface 12a thereof) and the second liquid cooling plate 14 (or the surface 14a thereof). Thereby, each of the first thermosiphon device 16 and the second thermosiphon device 18 is directly thermally coupled to the first liquid cooling plate 12 (or the surface 12a thereof) and the second liquid cooling plate 14 (or the surface 14a thereof). Through heat exchange between the first thermosiphon device 16 and the second thermosiphon device 18 with the first liquid cooling plate 12 and the second liquid cooling plate 14, the first liquid cooling plate 12 and the second liquid cooling plate 14 can absorb heat from the first thermosiphon device 16 and the second thermosiphon device 18, thereby achieving the heat dissipation effect. Furthermore, in the embodiment, the first liquid cooling plate 12 and the second liquid cooling plate 14 are connected through a communication structure 20. The communication structure 20 is located outside the first liquid cooling plate 12 and the second liquid cooling plate 14; that is, in the first direction D1, the projection of the communication structure 20 will not overlap with the projection of the first liquid cooling plate 12 and the second liquid cooling plate 14.
- [0016] Please refer to FIG. 4, which is a schematic diagram illustrating flow channels inside the first liquid cooling plate 12; the flow channels are shown in cross section. The first liquid cooling plate 12 has a first left side flow channel 122 (whose extent is shown by a frame in chain lines in the figure), a first right side flow channel 124 (whose extent is shown by a frame in chain lines in the figure), a first inlet 126, and a first outlet 128. The first left side flow channel 122 and the first right side flow channel 124 are connected in parallel to form a first parallel flow channel. Each of the first inlet 126 and the first outlet 128 is connected to both ends of the first parallel flow channel. In other words, the first inlet 126 is connected to one end of the first left side flow channel 122 and one end of the first right side flow channel 124 at the same time; the first outlet 128 is connected to the other end of the first left side flow channel 122 and the other end of the first right side flow channel 124 at the same time. In practical applications, the working fluid will flow into the first

liquid cooling plate **12** from the first inlet **126**, and flow separately through the first left side flow channel **122** and the first right side flow channel **124**, and finally flow out of the first liquid cooling plate **12** from the first outlet **128**. Therein, the flow direction of the working fluid in the first liquid cooling plate **12** is indicated by hollow arrows.

[0017] Furthermore, in the embodiment, each of the first left side flow channel **122** and the first right side flow channel **124** is a U-shaped flow channel and they are symmetrical in structure. Taking the first left side flow channel **122** as an example, the first left side flow channel **122** includes a first section **122a**, a second section **122b**, a third section **122c** and a fourth section **122d** (whose extents are shown by frames in dashed lines in the figure) in the order of the flow direction. The first section **122a** is opposite to the fourth section **122d**; the second section **122b** is opposite to the third section **122c**. The flow resistance of the first left side flow channel **122** between the first section **122a** and the second section **122b** is less than the flow resistance of the first section **122a** and the flow resistance of the second section **122b**. The flow resistance of the first left side flow channel **122** between the third section **122c** and the fourth section **122d** is less than the flow resistance of the third section **122c** and the flow resistance of the fourth section **122d**. This can be achieved by designing the cross-sectional size, shape, flow channel wall type, etc. of the first left side flow channel **122** at each section. This difference in flow resistance helps the working fluid mix between the sections (e.g., between the first section **122a** and the second section **122b**, between the third section **122c** and the fourth section **122d**), increases the uniformity of the temperature of the working fluid, and avoiding the generation of local hot spots.

[0018] In the embodiment, the first left side flow channel **122** includes three cavity sections **122e** and **122f** (whose extents are shown by frames in dashed lines in the figure). There is one cavity section **122e** between the first section **122a** and second section **122b**. There is another cavity section **122e** between the third section **122c** and the fourth section **122d**. There is one cavity section **122f** between the second section **122b** and the third section **122c**. There are fins in each of the first section **122a**, the second section **122b**, the third section **122c**, and the fourth section **122d**, while no fin is provided in the cavity sections **122e** and **122f**. This fin configuration can make the flow resistance of the first left side flow channel **122** in the cavity sections **122e** and **122f** less than the flow resistance of the first left side flow channel **122** in each of the first section **122a**, the second section **122b**, the third section **122c**, and the fourth section **122d**. The fins in the first left side flow channel **122** can increase the heat exchange efficiency.

[0019] Furthermore, the above descriptions about the first left side flow channel **122** also apply to the first right side flow channel **124**, and will not be repeated in addition. Furthermore, although the above descriptions of the first liquid cooling plate **12** take the structural symmetry of the first left side flow channel **122** and the first right side flow channel **124** as an example, the implementation is not limited thereto. In addition, as shown by FIG. 1 to FIG. 4, in the embodiment, there are a plurality of fins **13** which are disposed on a surface **12b** of the first liquid cooling plate **12** (opposite to the surface **12a**) and extend in a second direction D2 (indicated by a double-headed arrow in the figures). The fins **13** can also exchange heat with the surrounding environment of the first liquid cooling plate **12**. For example, when the cooling system **1** is installed in an equipment chassis, the chassis configuration can be designed so that the cooling airflow in the chassis (e.g., generated by fans) can flow through these fins **13** (e.g., the flow direction of the cooling airflow flowing here is roughly parallel to the second direction D2). Thereby, the first liquid cooling plate **12** can also cool the cooling airflow through these fins **13**, which can reduce the cooling load inside the chassis (e.g., reducing the operating power of the cooling fans), also reduce the cooling load outside the chassis (e.g., the operating power of the cooling fans in the cabinet accommodating multiple devices), and also help to reduce the air conditioning load of the computer room (accommodating the above equipment or cabinet).

[0020] Please refer to FIG. 5, which is a schematic diagram illustrating flow channels inside the second liquid cooling plate **14**; the flow channels are shown in cross section. The second liquid

cooling plate **14** has a second left side flow channel **142** (whose extent is shown by a frame in chain lines in the figure), a second right side flow channel **144** (whose extent is shown by a frame in chain lines in the figure), a second inlet **146**, and a second outlet **148**. The second left side flow channel **142** and the second right side flow channel **144** are connected in parallel to form a second parallel flow channel. Each of the second inlet **146** and the second outlet **148** is connected to both ends of the second parallel flow channel. In other words, the second inlet **146** is connected to one end of the second left side flow channel **142** and one end of the second right side flow channel **144** at the same time; the second outlet **148** is connected to the other end of the second left side flow channel **142** and the other end of the second right side flow channel **144** at the same time. In practical applications, the working fluid will flow into the second liquid cooling plate **14** from the second inlet **146**, and flow separately through the second left side flow channel **142** and the second right side flow channel **144**, and finally flow out of the second liquid cooling plate **14** from the second outlet **148**. Therein, the flow direction of the working fluid in the second liquid cooling plate **14** is indicated by hollow arrows.

[0021] Furthermore, in the embodiment, each of the second left side flow channel **142** and the second right side flow channel **144** is a U-shaped flow channel and they are symmetrical in structure. Taking the second left side flow channel **142** as an example, the second left side flow channel **142** includes a first section **142a**, a second section **142b**, a third section **142c** and a fourth section **142d** (whose extents are shown by frames in dashed lines in the figure) in the order of the flow direction. The first section **142a** is opposite to the fourth section **142d**; the second section **142b** is opposite to the third section **142c**. The flow resistance of the second left side flow channel **142** between the first section **142a** and the second section **142b** is less than the flow resistance of the first section **142a** and the flow resistance of the second section **142b**. The flow resistance of the second left side flow channel **142** between the third section **142c** and the fourth section **142d** is less than the flow resistance of the third section **142c** and the flow resistance of the fourth section **142d**. This can be achieved by designing the cross-sectional size, shape, flow channel wall type, etc. of the second left side flow channel **142** at each section. This difference in flow resistance helps the working fluid mix between the sections (e.g., between the first section **142a** and the second section **142b**, between the third section **142c** and the fourth section **142d**), increases the uniformity of the temperature of the working fluid, and avoiding the generation of local hot spots.

[0022] In the embodiment, the second left side flow channel **142** includes three cavity sections **142e** and **142f** (whose extents are shown by frames in dashed lines in the figure). There is one cavity section **142e** between the first section **142a** and second section **142b**. There is another cavity section **142e** between the third section **142c** and the fourth section **142d**. There is one cavity section **142f** between the second section **142b** and the third section **142c**. There are fins in each of the first section **142a**, the second section **142b**, the third section **142c**, and the fourth section **142d**, while no fin is provided in the cavity sections **142e** and **142f**. This fin configuration can make the flow resistance of the second left side flow channel **142** in the cavity sections **142e** and **142f** less than the flow resistance of the second left side flow channel **142** in each of the first section **142a**, the second section **142b**, the third section **142c**, and the fourth section **142d**. The fins in the second left side flow channel **142** can increase the heat exchange efficiency.

[0023] Furthermore, the above descriptions about the second left side flow channel **142** also apply to the second right side flow channel **144**, and will not be repeated in addition. Furthermore, although the above descriptions of the second liquid cooling plate **14** take the structural symmetry of the second left side flow channel **142** and the second right side flow channel **144** as an example, the implementation is not limited thereto. In addition, in practice, it is practicable to dispose a plurality of fins on a surface **14b** of the second liquid cooling plate **14** (opposite to the surface **14a**) to exchange heat with the surrounding environment, thereby producing essentially the same effect as the above fins **13**.

[0024] Furthermore, please refer to FIG. 1, FIG. 4 and FIG. 5. The first outlet **128** of the first liquid

cooling plate **12** and the second inlet **146** of the second liquid cooling plate **14** are connected through the communication structure **20**, so that the first parallel flow channel of the first liquid cooling plate **12** and the second parallel flow channel of the second liquid cooling plate **14** are connected in series (through the communication structure **20**). Thereby, the working fluid will first flow through the first liquid cooling plate **12** and then through the second liquid cooling plate **14**. In the embodiment, the first direction **D1** is parallel to the direction of gravity. The second liquid cooling plate **14** is located above the first liquid cooling plate **12**. Therefore, in principle, the temperature of the working fluid in the second liquid cooling plate **14** will be higher than the temperature of the working fluid in the first liquid cooling plate **12**, which helps to avoid or reduce the interference of the temperature gradient of the working fluid on the flow of the working fluid. [0025] In addition, in practice, the communication structure **20** can be a hose. In structural logic, the communication structure **20** may also be a combination of two channel structures arranged on the first liquid cooling plate **12** and the second liquid cooling plate **14**. Each channel structure extends from the corresponding first outlet **128** of the first liquid cooling plate **12** or the second inlet **146** of the second liquid cooling plate **14**. The two channel structures are then docked to form the communication structure **20**. In practice, each channel structure can be integrally formed with the corresponding first liquid cooling plate **12** or second liquid cooling plate **14**. Furthermore, in practice, the first inlet **126** and the second outlet **148** are connected to external pipelines (e.g., the manifold on the server cabinet, not shown in the figures). For example, the working fluid flowing through the first liquid cooling plate **12** and the second liquid cooling plate **14** flows to the external heat exchanger through these pipelines to dissipate heat.

[0026] Furthermore, as shown by FIG. 1 to FIG. 5, in the embodiment, the first thermosiphon device **16** corresponds to the first left side flow channel **122** and the second left side flow channel **142** and is thermally coupled between the first liquid cooling plate **12** and the second liquid cooling plate **14** (for example, the first thermosiphon device **16** directly contacts the surface **12a** of the first liquid cooling plate **12** and the surface **14a** of the second liquid cooling plate **14**, and the contact surfaces thereof can be coated with thermal conductive glue). The second thermosiphon device **18** corresponds to the first right side flow channel **124** and the second right side flow channel **144** and is thermally coupled between the first liquid cooling plate **12** and the second liquid cooling plate **14** (for example, the second thermosiphon device **18** directly contacts the surface **12a** of the first liquid cooling plate **12** and the surface **14a** of the second liquid cooling plate **14**, and the contact surfaces thereof can be coated with thermal conductive glue). Therein, the first thermosiphon device **16** includes a condenser **162**, an evaporator **164**, two transfer tubes **166** connecting the condenser **162** and the evaporator **164**, and the working fluid flowing in the above components. The second thermosiphon device **18** includes a condenser **182**, an evaporator **184**, two transfer tubes **186** connecting the condenser **182** and the evaporator **184**, and the working fluid flowing in the above components. The first thermosiphon device **16** is thermally coupled to the first liquid cooling plate **12** and the second liquid cooling plate **14** through the condenser **162**. The first thermosiphon device **16** is thermally coupled to a heat source (e. g., a processor) through the evaporator **164**. Thereby, the first thermosiphon device **16** can transfer the heat generated by the heat source to the first liquid cooling plate **12** and the second liquid cooling plate **14**, and then the heat can be dissipated through the external heat exchanger. The second thermosiphon device **18** is thermally coupled to the first liquid cooling plate **12** and the second liquid cooling plate **14** through the condenser **182**. The second thermosiphon device **18** is thermally coupled to a heat source (e.g., a processor) through the evaporator **184**. Thereby, the second thermosiphon device **18** can transfer the heat generated by the heat source to the first liquid cooling plate **12** and the second liquid cooling plate **14**, and then the heat can be dissipated through the external heat exchanger.

[0027] Furthermore, through the first parallel flow channel of the first liquid cooling plate **12** and the second parallel flow channel of the second liquid cooling plate **14**, and the series connection of the first parallel flow channel and the second parallel flow channel, the first liquid cooling plate **12**

and the second liquid cooling plate **14** provide similar cooling conditions for the two heat sources (e. g., processors) that are thermally coupled to the first thermosiphon device **16** and the second thermosiphon device **18** respectively, which can effectively reduce the temperature difference between the two heat sources.

[0028] Furthermore, through the series configuration of the first parallel flow channel of the first liquid cooling plate **12** and the second parallel flow channel of the second liquid cooling plate **14**, the number of connections between the liquid cooling plates and external piping (e.g., the manifold on the server cabinet, not shown in the figures) is relatively reduced, which can further reduces the obstruction of the external piping to the cooling airflow within the equipment chassis, thereby reducing fan power consumption.

[0029] Furthermore, as shown by FIG. 4 and FIG. 5, the fins in the first liquid cooling plate **12** and the second liquid cooling plate **14** are in principle thin plates and extend parallel to the flow direction of the working fluid. In practice, the fins in the first liquid cooling plate **12** and the second liquid cooling plate **14** can also be implemented by other types of fins; for example, the fin is a column (as shown by FIG. 6), a cone (as shown by FIG. 7), or a column with a straight cut (As shown by FIG. 8). Furthermore, in practice, the above fin structure (i.e., a structural configuration formed by the plurality of fins) can also be replaced by other structures that can also increase the contact area with the working fluid. For example, as shown by FIG. 9, a plurality of sheets **22** are used to form a structure that can increase the contact area with the working fluid (logically, it can also be regarded as a fin structure), thereby replacing the above fins disposed in the first liquid cooling plate **12** and the second liquid cooling plate **14**. Therein, the sheet **22** is bent to form a plurality of channels **222** which extend parallel to a third direction D3 (indicated by a double-headed arrow in the figure, e.g., parallel to the flow direction of the first left side flow channel **122**). The plurality of sheets **22** are arranged in the third direction D3. The channels **222** of the two sheets **22** that are adjacent disposed are misaligned.

[0030] In addition, the cooling system **1** can be used in servers, which can be used for artificial intelligence (AI) computing, edge computing, and can also be used as 5G servers, cloud servers or vehicle-to-everything servers.

[0031] Those skilled in the art will readily observe that numerous modifications and alterations of the device and method may be made while retaining the teachings of the invention. Accordingly, the above disclosure should be construed as limited only by the metes and bounds of the appended claims.

Claims

1. A cooling system, comprising: a first liquid cooling plate, the first liquid cooling plate having a first left side flow channel and a first right side flow channel, the first left side flow channel and the first right side flow channel being connected in parallel to form a first parallel flow channel; a second liquid cooling plate, the first liquid cooling plate and the second liquid cooling plate being spaced apart and opposite to each other, the second liquid cooling plate having a second left side flow channel and a second right side flow channel, the second left side flow channel and the second right side flow channel being connected in parallel to form a second parallel flow channel, the first parallel flow channel and the second parallel flow channel being connected in series; a first thermosiphon device, the first thermosiphon device being located between the first left side flow channel and the second left side flow channel and being thermally coupled to the first liquid cooling plate and the second liquid cooling plate; and a second thermosiphon device, the second thermosiphon device being located between the first right side flow channel and the second right side flow channel and being thermally coupled to the first liquid cooling plate and the second liquid cooling plate.

2. The cooling system according to claim 1, further comprising a communication structure, wherein

the communication structure is located outside the first liquid cooling plate and the second liquid cooling plate, the first liquid cooling plate has a first inlet and a first outlet, each of which is connected to both ends of the first parallel flow channel, the second liquid cooling plate has a second inlet and a second outlet, each of which is connected to both ends of the second parallel flow channel, and the first outlet and the second inlet are connected through the communication structure.

3. The cooling system according to claim 2, wherein the second liquid cooling plate is located above the first liquid cooling plate in a direction of gravity.

4. The cooling system according to claim 1, wherein the first left side flow channel and the first right side flow channel are both a U-shaped flow channel.

5. The cooling system according to claim 4, wherein the first left side flow channel comprises two sections in sequence, and a flow resistance of the first left side flow channel between the two sections is less than a flow resistance of either of the two sections.

6. The cooling system according to claim 5, wherein the first left side flow channel comprises a cavity section between the two sections, there are fins in the two sections, and there is no fin in the cavity section.

7. The cooling system according to claim 1, wherein the first left side flow channel and the first right side flow channel are symmetrical in structure.

8. The cooling system according to claim 1, wherein a plurality of fins are disposed in the first left side flow channel or the first right side flow channel, and the fin is a thin plate, a column, a cone, and a column with a straight cut.

9. The cooling system according to claim 1, wherein two sheets are disposed in the first left side flow channel, each sheet is bent to form a plurality of channels which extends parallel to a flow direction of the first left side flow channel, the two sheets are adjacently arranged in the flow direction, and the channels of the two sheet are misaligned.

10. The cooling system according to claim 1, wherein the first thermosiphon device comprises a condenser, an evaporator, and two transfer tubes, each transfer tube is connected to the condenser and the evaporator, and the first thermosiphon device is thermally coupled to the first liquid cooling plate and the second liquid cooling plate through the condenser.
